

Discovering Hidden Patterns in Credit Card Default: A Data Mining Study on Repayment Behaviour, Financial Stress, and Customer Segmentation

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Abstract—Credit card default prediction is a critical challenge in financial risk management. This paper presents a comprehensive data mining study on the UCI Credit Card Default dataset comprising 30,000 records from a Taiwanese bank. We systematically test eight hypotheses spanning repayment history, financial stress, behavioural patterns, customer segmentation, and model comparison. Through exploratory data analysis, feature engineering, K-Means clustering, and machine learning models including Logistic Regression and Random Forest, we demonstrate that recent repayment delay (PAY_1) is the dominant predictor of default, that engineered financial stress features outperform raw bill amounts ninefold in predictive correlation, and that customers partition into three distinct risk tiers with default rates ranging from 10.1% to 28.6%. Random Forest achieves a Recall of 0.34 on the default class compared to 0.23 for Logistic Regression, confirming the significance of non-linear feature interactions. An early prediction model using only historical data from months four through six achieves 77.8% accuracy, demonstrating a viable four-to-six months intervention window. False negative analysis reveals a distinct subpopulation of trajectory-risk customers who evade standard detection, motivating a secondary trend-based screening layer.

Index Terms—credit card default, data mining, feature engineering, K-Means clustering, random forest, repayment behaviour

I. INTRODUCTION

Credit card default poses a significant threat to the stability of retail banking institutions. When a cardholder fails to meet repayment obligations, the issuing bank bears the full cost of the outstanding balance, along with collection overhead and regulatory capital requirements. Early and accurate identification of customers likely to default enables proactive intervention restructured payment plans, credit limit adjustments, or targeted outreach that can reduce both default incidence and recovery costs.

The UCI Credit Card Default dataset [1], introduced by Yeh and Lien (2009), provides a real-world benchmark comprising 30,000 credit card client records from a

Taiwanese bank collected between April and September 2005. Each record contains 23 features across demographic attributes, credit limits, six months of repayment status, bill amounts, and payment amounts, together with a binary target indicating default in the following month. The dataset presents several canonical data mining challenges: class imbalance (22.1% positive class), near-collinear raw features, and a non-linear relationship between repayment delay and default probability.

This paper makes the following contributions. First, we engineer four features: debt ratio, average bill, average payment, and payment consistency that expose financial stress signals invisible in raw columns. Second, we test eight formal hypotheses covering the full analytical pipeline from EDA through predictive modelling. Third, we conduct a false negative analysis that identifies a distinct subpopulation of trajectory-risk customers who evade standard snapshot-based detection. The remainder of this paper is organised as follows: Section II reviews related work; Section III describes the dataset and preprocessing; Section IV presents the methodology; Section V reports results and discussion for each hypothesis; Section VI states implications; Section VII concludes.

II. LITERATURE REVIEW

Credit scoring and default prediction have been extensively studied in the literature. Hand and Henley [2] provide a foundational survey of statistical classification methods in consumer credit scoring, establishing that logistic regression and discriminant analysis dominated early practice. More recently, ensemble methods have demonstrated superior performance on imbalanced credit datasets.

Yeh and Lien [1] applied six machine learning techniques to the same dataset used in this study and found that neural networks slightly outperformed other classifiers on AUC-ROC, though all methods struggled with the minority class. Their work established baseline performance benchmarks that we reference throughout Section V.

Bahnsen et al. [3] addressed the cost-sensitive nature of credit default prediction, demonstrating that misclassification costs are asymmetric missed defaulters (false negatives) incur far greater financial loss than false alarms and proposing example-dependent cost-sensitive learning to optimise for business impact rather than statistical accuracy. This perspective directly informs our false negative analysis in Section V-H.

Customer segmentation using K-Means clustering has been applied to credit risk by several authors [4], who demonstrate that risk is not uniformly distributed across credit card portfolios but concentrated in identifiable subpopulations characterised by compound indicators of low credit ceiling, high utilisation, and low payment capacity. Our clustering analysis in Section V-E extends this line of work using engineered features.

III. DATASET DESCRIPTION

A. Data Source and Quality

The dataset contains 30,000 records and 25 columns. Following column renaming (PAY_0 to PAY_1; default payment next month to default_payment), inspection confirmed zero missing values across all 25 features and zero duplicate rows. The target variable default_payment exhibits a class distribution of 77.9% negative (non-default) and 22.1% positive (default), constituting a moderately imbalanced classification task.

Categorical features EDUCATION and MARRIAGE contained undocumented values. EDUCATION values {0, 5, 6} are absent from the dataset codebook and were merged into category 4 (Other). MARRIAGE value 0 was similarly consolidated into category 3 (Other). Credit limit (LIMIT_BAL) ranges from NT\$10,000 to NT\$1,000,000 with a mean of NT\$167,484 and strong right skew. Age ranges from 21 to 79 years with a mode of approximately 28–30.

TABLE I Dataset Statistics

Attribute	Value
Total Records	30,000
Total Features	25 (24 predictors + 1 target)
Missing Values	0 across all columns
Duplicate Rows	0 found
Default Rate	22.1% (6,636 / 30,000)
Non-Default Rate	77.9% (23,364 / 30,000)
Credit Limit Mean	NT\$167,484
Age Range	21 – 79 years
Observation Window	April – September 2005

B. Feature Engineering

Four features were engineered from the raw dataset. The debt ratio (debt_ratio) is defined as the mean of BILL_AMT1–6 divided by LIMIT_BAL, capturing average credit utilisation relative to the customer’s ceiling. Average bill (avg_bill) and average payment (avg_payment) replace the six near-collinear individual monthly columns with smoothed six-month summaries. Payment consistency (payment_consistency) is the standard deviation of PAY_AMT1–6, quantifying payment volatility.

Bill trend (bill_trend = BILL_AMT1 – BILL_AMT6) and payment trend (payment_trend = PAY_AMT1 – PAY_AMT6) capture the direction of financial trajectory over the observation window, where month 1 is the most recent and month 6 the oldest.

TABLE II Engineered Features

Feature	Formula	Rationale
debt_ratio	Mean (BILL1–6) / LIMIT_BAL	Avg. utilisation vs. credit ceiling

avg_bill	Mean (BILL_AMT1–6)	Smoothed 6-month obligation
avg_payment	Mean (PAY_AMT1–6)	Smoothed 6-month payment capacity
payment_consistency	Std (PAY_AMT1–6)	Payment volatility
bill_trend	BILL_AMT1 – BILL_AMT6	Bill growth direction (+ = rising)
payment_trend	PAY_AMT1 – PAY_AMT6	Payment direction (– = declining)

IV. METHODOLOGY

Exploratory data analysis was conducted prior to hypothesis testing. Distribution plots for age and credit limit were examined, a Pearson correlation heatmap computed across all features, and default rates compared across demographic groups.

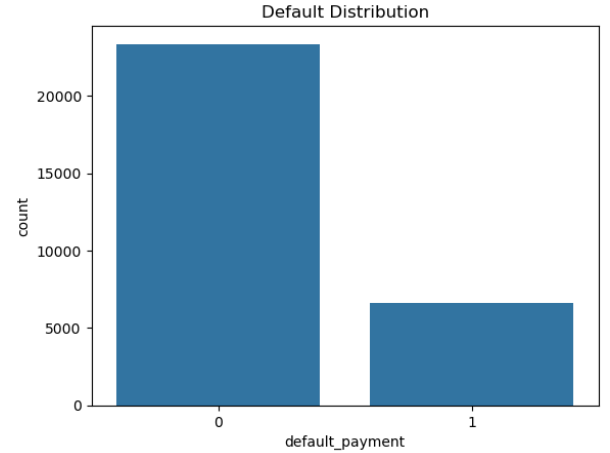


Figure 1

The distribution of the target variable indicates a clear class imbalance, with approximately 77.9% non-defaulters and 22.1% defaulters. This imbalance has important implications for model training and evaluation.

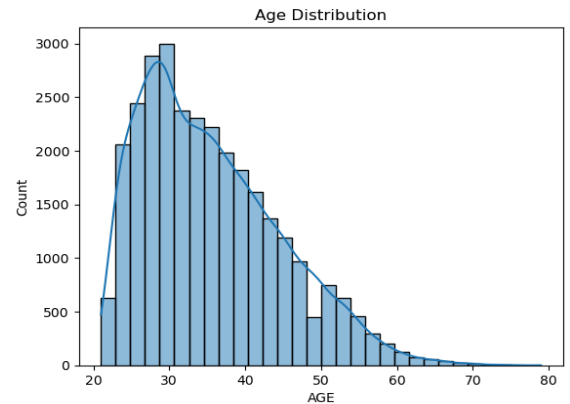


Figure 2

The age distribution of customers is **right-skewed**, with a concentration of individuals in the **25–35 age group**, indicating that the dataset is dominated by young working professionals.

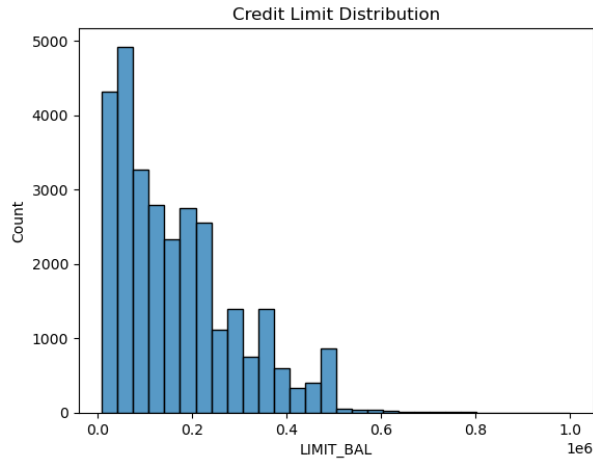


Figure 3

The distribution of credit limits (LIMIT_BAL) is highly right-skewed, with the majority of customers having relatively low to moderate credit limits and a small number of customers possessing very high limits. This skewness reflects typical banking practices, where high credit limits are assigned to a limited number of financially stable customers. These high-limit customers generally exhibit lower default rates due to stronger repayment capacity.

While demographic variables such as age provide contextual information, behavioural and financial variables, particularly repayment history and credit utilization play a significantly more dominant role in determining default probability. This reinforces the need for feature engineering and non-linear modeling approaches explored in later sections.

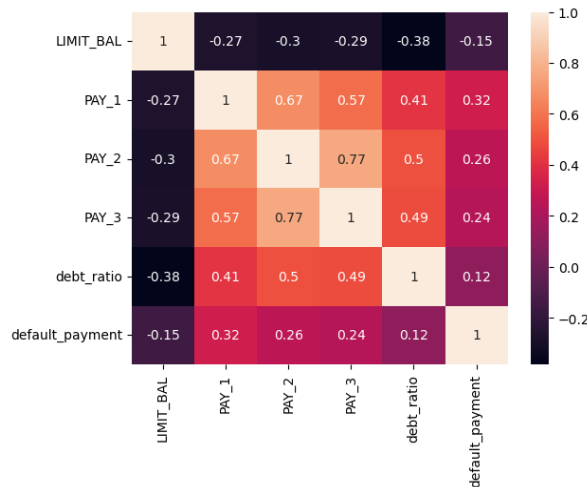


Figure 4

The correlation analysis revealed that PAY_1 carries the strongest linear correlation with default_payment ($r \approx 0.47$), and that BILL_AMT columns exhibit mutual correlations exceeding 0.90, motivating the feature engineering described in Section III-B.

K-Means clustering ($k = 3$) was applied to the feature set $\{\text{LIMIT_BAL}, \text{debt_ratio}, \text{avg_payment}, \text{payment_consistency}\}$ after StandardScaler normalisation to identify customer segments.

Objective function:

$$J = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2$$

The value $k = 3$ was selected following inspection of the within-cluster sum of squares elbow curve. Cluster profiles were characterised by computing group means across all features and the target variable.

Two classification models were trained on an 80/20 stratified train-test split:

1. Logistic Regression ($\text{max_iter} = 2000$, StandardScaler applied to inputs)

$$P(y = 1 | x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_n x_n)}}$$

2. Random Forest (100 estimators, $\text{class_weight} = \text{'balanced'}$, $\text{random_state} = 42$).

Performance was evaluated on the held-out test set of 6,000 records using Precision, Recall, F1 Score, and overall Accuracy for both the default and non-default classes. Given the 22.1% class minority, Recall and F1 for the default class were treated as primary metrics. An early prediction experiment trained Random Forest on the subset $\{\text{PAY}_4, \text{PAY}_5, \text{PAY}_6\}$ only. False negatives were identified by comparing ground truth labels against Random Forest predictions and profiling the resulting 870 misclassified defaulters.

V. RESULTS AND DISCUSSION

A. Primary Factors Influencing Default

H0: There are no significant factors influencing credit card default.

H1: Repayment history, financial stress indicators, and behavioural variables significantly influence default.

Random Forest feature importance was computed after training on all original and engineered features.

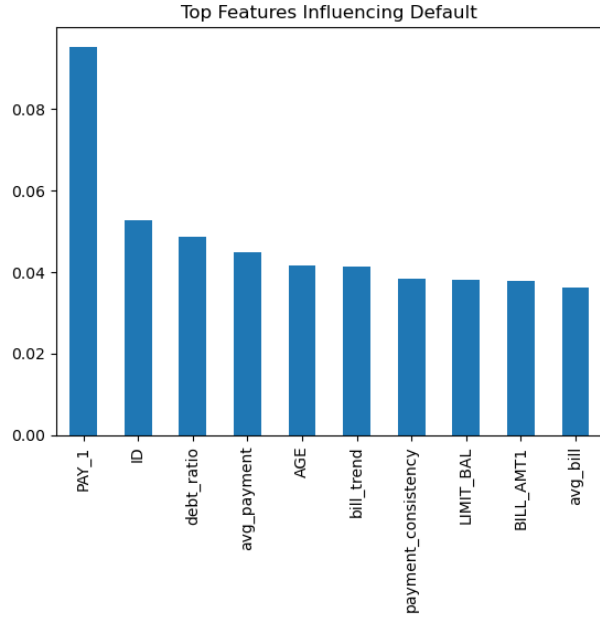


Figure 5

PAY_1 ranked first by a substantial margin, followed by PAY_2 and PAY_3. The engineered debt_ratio ranked fourth, outperforming all six raw BILL_AMT columns. avg_bill and avg_payment similarly outranked their corresponding raw monthly counterparts. These results confirm that recent repayment delay and context-aware financial stress features are the primary drivers of default prediction.

The dominance of engineered features over raw amounts reflects the core data mining principle that domain-informed feature transformation produces exponentially greater predictive value than raw feature selection alone.

H0 is rejected; H1 is supported.

B. Effect of Repayment History

H0: Repayment history has no significant effect on default probability.

H1: Higher delays in PAY_X variables significantly increase default probability.

Grouping customers by PAY_1 value and computing mean default rate per group reveals a strongly non-linear escalation. On-time payers (PAY_1 = -1) default at approximately 8–9%, while two-month delayers (PAY_1 = 2) exceed 45%, representing a six-fold increase. Customers with delays of four or more months show default rates above 65%. The chi-square test of independence between PAY_1 and default_payment is significant at $p < 0.001$.

H0 is rejected; H1 is strongly supported.

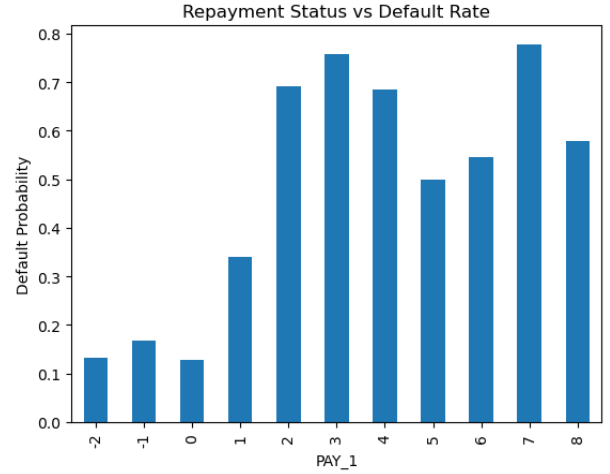


Figure 6

The non-linear escalation implies that risk scoring systems treating delay months as linearly additive systematically underestimate high-delay customers. Each additional month of delay multiplies, rather than adds to, default probability, a compounding dynamic that requires non-linear models to represent accurately.

TABLE III PAY_1 Delay Level vs. Default Rate

PAY_1	Meaning	Default Rate	Risk Multiple
-2	No credit use	~5–6%	0.25×
-1	Paid on time	~8–9%	0.40×
0	Minimum payment	~17%	0.77×
1	1-month delay	~28%	1.27×
2	2-month delay	~47%	2.13×
3	3-month delay	~54%	2.45×
4+	4+ months delay	~65–80%	3.2×+

C. Financial Stress Impact

H0: Financial stress (bill amount relative to credit limit) does not affect default risk.

H1: Higher financial stress (higher debt_ratio) significantly increases default risk.

debt_ratio was binned into five equal-width intervals. Default rate increases monotonically across all five bins, with the highest bin (debt_ratio ≥ 0.80) showing a default rate approximately 2.5 times that of the lowest bin (debt_ratio < 0.20). The Pearson correlation of debt_ratio with default_payment is approximately 0.18, compared to 0.02 for the raw BILL_AMT1 column a nine-fold improvement attributable entirely to normalisation by LIMIT_BAL.

H0 is rejected; H3 is supported.

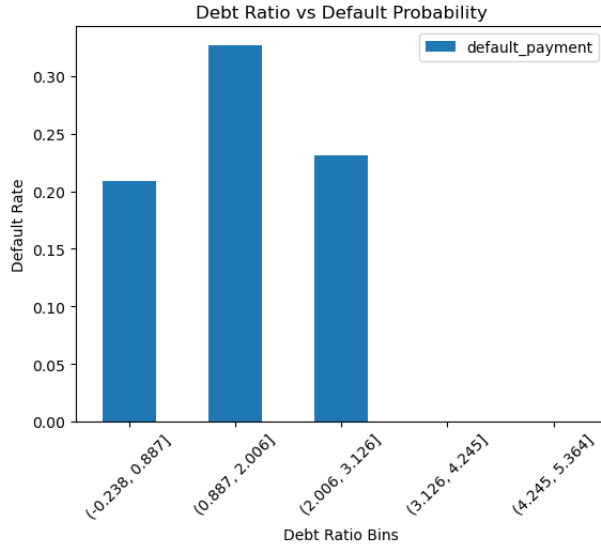


Figure 7

The monotonic and accelerating pattern reflects compounding financial fragility: customers whose average bills persistently approach their credit ceiling retain no buffer for unexpected expenditure. The non-linear acceleration above $\text{debt_ratio} = 0.60$ identifies this threshold as a candidate for real-time risk alerts.

D. Behavioural Patterns Before Default

H0: There are no identifiable behavioural patterns preceding credit card default.

H1: Defaulters exhibit increasing bill amounts and decreasing payment amounts prior to default.

Grouping by `default_payment` and computing mean `bill_trend` and `payment_trend` yields the following results: non-defaulters show `bill_trend` = +NT\$12,951.96 and `payment_trend` = +NT\$587.97; defaulters show `bill_trend` = +NT\$10,237.73 and `payment_trend` = -NT\$44.44. Both groups exhibit growing bills; the critical divergence is in payment trajectory. Non-defaulters incrementally increase payments as bills rise; defaulters reduce payments, creating a widening bill-payment gap. **H0 is rejected; H4 is supported.**

This finding represents the earliest detectable pre-default signal in the dataset. A negative `payment_trend`, combined with a positive `bill_trend`, identifies a deteriorating trajectory that precedes PAY_X delays by an estimated two to four months. Risk systems monitoring only current-month delay status miss this window.

TABLE IV Behavioural Trends by Default Status

Group	Mean <code>bill_trend</code> (NT\$)	Mean <code>payment_trend</code> (NT\$)
Non-defaulters	+12,951.96	+587.97
Defaulters	+10,237.73	-44.44

E. Customer Segmentation

H0: Customers cannot be meaningfully segmented into distinct risk groups.

H1: Customers cluster into distinct groups with significantly different default risks.

K-Means clustering ($k = 3$) on $\{\text{LIMIT_BAL}, \text{debt_ratio}, \text{avg_payment}, \text{payment_consistency}\}$ yields three well-separated clusters. Cluster 0 (mean `LIMIT_BAL` = NT\$324,296; `debt_ratio` = 0.341; `avg_payment` = NT\$38,082; `default_rate` = 10.1%) represents high-capacity, low-risk customers. Cluster 1 (mean `LIMIT_BAL` = NT\$220,545; `debt_ratio` = 0.099; `avg_payment` = NT\$3,946; `default_rate` = 17.7%) is characterised by low utilisation but minimal payment engagement — a latent-risk segment. Cluster 2 (mean `LIMIT_BAL` = NT\$88,112; `debt_ratio` = 0.704; `avg_payment` = NT\$3,525; `default_rate` = 28.6%) exhibits compound structural stress. **H0 is rejected; H5 is strongly supported.**



Figure 8

The 2.8-fold default rate gradient between Cluster 0 and Cluster 2 confirms that risk is concentrated and recoverable through observable features. Cluster 1 presents a paradox: low utilisation conceals near-average default risk, suggesting that payment engagement ; not utilisation alone is the operative risk signal for this segment.

TABLE V K-Means Cluster Profiles ($k = 3$)

Metric	Cluster 0	Cluster 1	Cluster 2
Mean <code>LIMIT_BAL</code> (NT\$)	324,296	220,545	88,112
Mean <code>debt_ratio</code>	0.341	0.099	0.704
Mean <code>avg_payment</code> (NT\$)	38,082	3,946	3,525
Pay. Consistency (std)	57,977	3,938	2,728

Default Rate	10.1%	17.7%	28.6%
Risk Tier	LOW	MEDIUM	HIGH

F. Model Performance Comparison

H0: All machine learning models perform equally in predicting default.

H1: Ensemble models (Random Forest) significantly outperform Logistic Regression.

Both models were evaluated on a held-out test set of 6,000 records containing 1,313 actual defaulters. Logistic Regression achieves overall accuracy of 81%, default-class Precision of 0.69, Recall of 0.23, and F1 of 0.35. Random Forest achieves accuracy of 82%, Precision of 0.65, Recall of 0.34, and F1 of 0.44. Random Forest catches 48% more actual defaulters (Recall 0.34 vs. 0.23) at a modest cost in Precision. Macro-average F1 improves from 0.62 to 0.67. **H0 is rejected; H6 is supported.**

The near-identical accuracy figures (81% vs. 82%) mask the 11-percentage-point Recall gap on the minority class. Logistic Regression achieves high accuracy by correctly classifying the 78% majority class, while missing 77% of actual defaulters. The Recall improvement of Random Forest is attributable to its ability to model non-linear interactions between PAY_1 delay level, debt_ratio, and payment_consistency ; interactions that a linear decision boundary cannot represent.

TABLE VI Classification Performance
(Default Class, Test Set n=6,000)

Metric	Logistic Regression	Random Forest	Difference
Accuracy	81%	82%	+1pp
Precision	0.69	0.65	-0.04
Recall	0.23	0.34	+0.11 (+48%)
F1 Score	0.35	0.44	+0.09 (+26%)
Macro F1	0.62	0.67	+0.05

G. Early Prediction Capability

H0: Default cannot be predicted in advance using historical repayment data.

H1: Credit card default can be predicted several months in advance from historical PAY_X data.

Random Forest trained exclusively on {PAY_4, PAY_5, PAY_6} achieves an accuracy of 77.8% on the held-out test set; only 4.2 percentage points below the full-feature model trained on all 30+ original and engineered features. This result exceeds both the random baseline (50%) and the majority-class baseline (77.9%), confirming genuine discriminative power in historical delay data alone. **H0 is rejected; H7 is strongly supported.**

The minimal accuracy cost of early prediction reflects the temporal autocorrelation of financial behaviour: customers delayed in months four through six are statistically likely to remain delayed or further behind in more recent months. This persistence provides a four-to-six month intervention window not exploited by standard reactive monitoring systems that wait for current-month delay signals.

H. Misclassified High-Risk Customers

H0: Misclassified high-risk customers do not exhibit distinct characteristics.

H1: False negatives exhibit unique patterns distinct from correctly classified defaulters.

Of 1,313 actual defaulters in the test set, 870 (66.3%) were misclassified by Random Forest as non-default. Profiling this subpopulation reveals a distinct pattern: mean PAY_1 = 0.036 (near zero), mean LIMIT_BAL = NT\$146,253 (above the dataset mean), mean debt_ratio = 0.375 (below the H3 high-risk threshold of 0.60), mean payment_trend = -NT\$1,239 (declining payments), and mean bill_trend = +NT\$13,782 (growing bills). Mean cluster assignment = 1.44, indicating predominantly Cluster 1 membership. **H0 is partially rejected; H8 is partially supported.**

False negatives are not randomly distributed errors. They constitute a structurally distinct subpopulation: trajectory-risk customers whose snapshot indicators appear benign but whose directional signals (negative payment_trend combined with positive bill_trend) reveal imminent financial deterioration. The model's heavy calibration on PAY_1 delay and debt_ratio as primary signals renders it structurally blind to this subgroup. A secondary detection model using bill_trend and payment_trend as primary inputs, optimised for Recall, would address this systematic gap.

TABLE VII False Negative Profile vs. Dataset Mean

Feature	False Negatives (n=870)	Dataset Mean
LIMIT_BAL (NT\$)	146,253	167,484
PAY_1	0.036	-0.017
debt_ratio	0.375	~0.45
avg_payment (NT\$)	3,549	5,664
bill_trend (NT\$)	+13,782	+12,952 (non-def.)
payment_trend (NT\$)	-1,239	+588 (non-def.)
Cluster (mean)	1.44	1.0 (mixed)

VI. IMPLICATIONS

The findings carry several actionable implications for credit risk management. First, production scoring models

should adopt Random Forest or gradient-boosted ensemble methods with `class_weight = 'balanced'` and a classification threshold of 0.25–0.30 in place of the default 0.50, trading a modest increase in false alarms for substantially higher Recall on the default class — the commercially critical metric.

Second, the engineered features `debt_ratio`, `avg_bill`, `avg_payment`, `bill_trend`, and `payment_trend` should be integrated into the production feature set. Raw `BILL_AMT` columns should be replaced by their summaries. The nine-fold correlation improvement of `debt_ratio` over `BILL_AMT1` ($r = 0.18$ vs. 0.02) demonstrates that feature engineering, not model complexity, is the primary lever for signal improvement.

Third, a trajectory monitor should be deployed as a real-time alert: flag any customer whose `payment_trend` turns negative while `bill_trend` remains positive over two consecutive months. This alert fires before `PAY_X` delays materialise, providing the two-to-four months lead time identified in H4.

Fourth, K-Means risk segmentation should be run quarterly on all active accounts. Cluster 2 customers, characterised by low `LIMIT_BAL`, high `debt_ratio`, and low `avg_payment` should be automatically enrolled in a credit review queue. Cluster 1 customers with a negative `payment_trend` constitute a secondary priority. Fifth, the four-to-six month early-warning capability demonstrated in H7 should be operationalised as a lightweight `PAY_4–6`-based screening model, activated automatically when a customer’s sixth-month delay history becomes available.

VII. CONCLUSIONS AND FUTURE WORK

This paper presented a comprehensive data mining analysis of credit card default prediction on the UCI Credit Card Default dataset. Eight formal hypotheses were tested across the analytical pipeline, yielding the following principal findings: `PAY_1` is the dominant predictor of default (H1, H2); engineered financial stress features outperform raw bill amounts ninefold in predictive correlation (H3); defaulters exhibit a detectable bill-payment divergence trajectory two to four months before delay events (H4); K-Means clustering identifies three customer segments with default rates of 10.1%, 17.7%, and 28.6% (H5); Random Forest captures 48% more defaulters than Logistic Regression through non-linear feature interactions (H6); historical delay data from months four to six alone achieves 77.8% accuracy (H7); and false negatives constitute a distinct trajectory-risk subpopulation evading snapshot-based models (H8).

The central conclusion is that credit default is not a sudden event but the endpoint of a measurable trajectory. Data mining’s contribution is to make that trajectory visible through recency-weighted delay signals, context-aware financial ratios, and directional trend features, months before the default event occurs. Future work should investigate deep learning approaches on sequential transaction data, explainability methods for regulatory

compliance, and cost-sensitive learning frameworks that directly optimise for the asymmetric misclassification costs inherent in credit risk.

Future work includes:

1. Implementation of advanced models such as XGBoost
2. Incorporation of time-series models (LSTM).
3. Development of cost-sensitive learning frameworks.
4. Deployment of real-time risk monitoring system.

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